Fixed-Centre Congruences and Complete Finite-Gap Shifted-Triple Exclusion for Nörlund–Stirling Quotients

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Abstract

For odd $n \geq 5$, let $S_n(X) = [t^n](t/(1-e^{-t}))^X$, proportional to a primitive integer Nörlund–Stirling quotient polynomial $q_n(X)$. We derive an exact prime-adic criterion for three equally spaced odd indices to vanish at $X = -3$ modulo a prime power. At good primes, a triple zero modulo p^3 forces p to be Wieferich simultaneously to bases 2 and 3. This implication is independent of the index gap. An exhaustive modular computation verifies that no prime $p \leq 10^7$ meets both necessary conditions, excluding all such fixed-centre triples at precision p^3 in that prime range. We also give a general interpolation bound at any negative integer centre. We give an exact necessary-and-sufficient test for the first shifted Hensel digit of such a triple, using its affine first-precision values and first derivative vector. For triples with a common residue at -3, we obtain a necessary first-order shifted-root curvature congruence and finite-prime restrictions for fixed index gaps; the $p = 431$ configuration fails this curvature test. Combining this with a complete all-prime gap certificate, we exclude all shifted three-order common roots modulo p^2 in the -3 disk for every even gap $2 \leq h \leq 120$, with no upper bound on the prime.

Contributions and novelty

The paper establishes five related results for Nörlund–Stirling quotient polynomials, with their hypotheses kept separate. First, it derives an exact prime-power criterion and divisibility depth for three equally spaced odd indices at the fixed centre $X = -3$, including propagation along the index progression. Second, it proves a two-base Wieferich necessary condition for triple depth at least three and its prime-bounded exclusion for $p \leq 10^7$. Third, it obtains an interpolation obstruction and an explicit bound on eligible primes and depth for fixed negative integer centres. Fourth, it gives an exact first-Hensel-digit compatibility criterion for a shifted triple in the specified residue disk. Fifth, a finite arithmetic certificate combined with the first-digit obstruction rules out all such triple roots modulo p^2 for every even gap $2 \leq h \leq 120$ and *every* prime satisfying the stated good-prime inequality.

The finite-difference identities, lifting-the-exponent arguments, Hensel expansion, and finite-field linear algebra themselves are classical. The contribution is the specified collection of exact congruence and exclusion results, not a new general method of Hensel lifting or an assertion of historical priority. The finite-gap theorem uses an explicit finite computational certificate as part of its proof; the prime-bounded Wieferich sieve is a separate calculation. The conclusions do not extend to arbitrary residue disks, unrestricted gaps, or pairs of orders.

1 Notation and exact negative-integer coefficients

For odd $n \geq 5$ set

$$S_n(X) = [t^n] \left(\frac{t}{1-e^{-t}} \right)^X = \frac{X^2(X-1)}{D_n} q_n(X),$$

where D_n is the canonical integer normalization of the primitive quotient polynomial q_n. The required good-prime normalization property is proved below, in the slightly stronger range $p > n+1$. Since $p \geq 5$ makes $(-3)^2(-4)$ a unit, zeros of $q_n(-3)$ and $S_n(-3)$ modulo p^k agree at each good prime.

Proposition 1 (Integral normalization at good primes)**.** *For odd $n \geq 5$, write*

$$S_n(X) = \frac{X^2(X-1)}{D_n} q_n(X),$$

where $q_n \in \mathbb{Z}[X]$ is primitive and D_n is the positive integer normalizing denominator. For every prime $p > n+1$,

$$p \nmid D_n.$$

More precisely, the polynomial $S_n(X)/(X^2(X-1))$ belongs to $\mathbb{Z}_{(p)}[X]$ and has leading coefficient $1/(2^n n!)$, a p-adic unit.

Proof. Set $H(t) = t/(1-e^{-t})$. Its reciprocal is

$$H(t)^{-1} = \sum_{j \geq 0} \frac{(-1)^j}{(j+1)!} t^j.$$

For $p > n+1$ all denominators up to degree n are p-adic units, so $H(t) \in \mathbb{Z}_{(p)}[[t]]$ through degree n. The binomial expansion

$$H(t)^X = \sum_{j=0}^{n} \binom{X}{j} (H(t)-1)^j + O(t^{n+1})$$

therefore proves $S_n(X) \in \mathbb{Z}_{(p)}[X]$, because $j!$ is a p-unit for $j \leq n$.

The identities $H(-t) = H(t) - t$ and $\log H(-t) = \log H(t) - t$ show that both $H(t) - t/2$ and $\log H(t) - t/2$ are even power series. Hence for odd $n \geq 5$,

$$S_n(0) = 0, \quad S_n'(0) = [t^n] \log H(t) = 0, \quad S_n(1) = [t^n] H(t) = 0.$$

Thus $X^2(X-1)$ divides S_n over $\mathbb{Q}[X]$. Since it is monic, polynomial division over $\mathbb{Z}_{(p)}[X]$ shows that the quotient is p-integral. Finally $\log H(t) = t/2 + O(t^2)$, so the coefficient of X^n in S_n (and the leading coefficient of the quotient) is

$$\frac{1}{n!} [t^n] (\log H(t))^n = \frac{1}{2^n n!},$$

a p-unit. If p divided D_n, every coefficient of the integer polynomial $q_n = D_n S_n/(X^2(X-1))$ would be divisible by p, contradicting primitiveness. Therefore $p \nmid D_n$. □

For $m \geq 2$, expansion of $((1-e^{-t})/t)^m$ gives the exact identity

$$-(n+m)! S_n(-m) = \sum_{a=1}^{m} (-1)^{m-a} \binom{m}{a} a^{n+m}, \quad n \geq 5 \text{ odd}. \tag{1}$$

Take even $E \geq 6$, even $h \geq 2$, and $n_j = E + jh - 1$. Equation (1) makes the dependence on the progression variable j a finite exponential sum.

2 An exact criterion for three odd orders at -3

Put

$$C_j = 3 \cdot 3^{E+jh} - 4 \cdot 2^{E+jh} + 1 \quad (j \geq 0), \qquad L = 4 \cdot 2^E,\ M = 3 \cdot 3^E,\ x = 2^h - 1,\ y = 3^h - 1.$$

The normalization in (1) gives $-(n_j+3)!S_{n_j}(-3) = 3C_j$. Write

$$D_1 = C_1 - C_0 = My - Lx, \qquad D_2 = C_2 - 2C_1 + C_0 = My^2 - Lx^2, \qquad C_0 = M - L + 1.$$

A direct expansion yields the identity

$$MD_2 = D_1^2 + 2LxD_1 + L(1 - C_0)x^2. \tag{2}$$

Theorem 2 (Exact fixed-centre triple criterion)**.** *For every prime $p \geq 5$, integers $E \geq 6$ and $h \geq 2$ even, and $k \geq 1$, the congruences $C_0 \equiv C_1 \equiv C_2 \equiv 0 \pmod{p^k}$ are equivalent to*

$$C_0 \equiv D_1 \equiv 0 \pmod{p^k}, \qquad (2^h - 1)^2 \equiv 0 \pmod{p^k}.$$

Equivalently, in valuation form,

$$K_p(E,h) = \min\{v_p(C_0), v_p(D_1), 2v_p(2^h - 1)\} \tag{3}$$

is the exact simultaneous divisibility depth of C_0, C_1, C_2. Moreover, divisibility modulo p^k forces $p^{\lceil k/2 \rceil}$ to divide both $2^h - 1$ and $3^h - 1$.

Proof. The triple congruence is equivalent to $C_0 \equiv D_1 \equiv D_2 \equiv 0 \pmod{p^k}$. Under $C_0 \equiv 0 \pmod{p}$, both L and M are p-adic units, as is $1 - C_0$. Formula (2) makes $D_2 \equiv 0$ equivalent to $x^2 \equiv 0 \pmod{p^k}$ if $D_1 \equiv 0$. This proves the iff statement and (3). Finally $My \equiv Lx \pmod{p^k}$ transfers the lower valuation bound from x to y. □

Proposition 3 (Propagation along the entire index progression)**.** *In the hypotheses of Theorem 2, if $C_0 = C_1 = C_2 = 0 \pmod{p^k}$, then $C_j = 0 \pmod{p^k}$ for every $j \geq 0$. Consequently, whenever the good-prime inequality $p > 2(E + jh - 1)$ holds, the corresponding primitive quotient q_{E+jh-1} also vanishes at $X = -3$ modulo p^k.*

Proof. Put $x = 2^h - 1$ and $y = 3^h - 1$. Theorem 2 gives $x^2 = y^2 = 0$ modulo p^k, and $C_0 = D_1 = 0$ in the same ring. The binomial identity yields, for each integer $j \geq 0$,

$$C_j = M(1+y)^j - L(1+x)^j + 1 \equiv C_0 + j(My - Lx) = C_0 + jD_1 = 0 \pmod{p^k}.$$

The final claim follows from the stated integral normalization and the factorial-unit condition at a good prime. □

3 The two-base Wieferich obstruction

Theorem 4. *Let $E \geq 6$ and $h \geq 2$ be even, and let $p > 2(E + 2h - 1)$ be prime. If the three quotient polynomials q_{E-1}, q_{E+h-1}, q_{E+2h-1} vanish at $X = -3$ modulo p^k, then for $b = 2$ and $b = 3$,*

$$v_p(b^{p-1} - 1) \geq \lceil k/2 \rceil.$$

In particular, if $k \geq 3$ then

$$2^{p-1} \equiv 3^{p-1} \equiv 1 \pmod{p^2}. \tag{4}$$

Proof. By Theorem 2, $p^{\lceil k/2 \rceil} \mid b^h - 1$ for each $b \in \{2, 3\}$. The good-prime inequality implies $h < p$. Set $d = \mathrm{ord}_p(b)$. Then $d \mid h$ and $d \mid p-1$, while p divides neither h/d nor $(p-1)/d$. The lifting-the-exponent lemma gives

$$v_p(b^h - 1) = v_p(b^d - 1) = v_p(b^{p-1} - 1).$$

The claim follows. □

Corollary 5. *For all even $E \geq 6$, even $h \geq 2$, and prime $2(E + 2h - 1) < p \leq 10^7$, the three quotient polynomials with indices $E - 1, E + h - 1, E + 2h - 1$ cannot all vanish at $X = -3$ modulo p^3.*

Proof. A complete deterministic sieve lists the 664,577 primes $5 \leq p \leq 10^7$. Among these, the primes satisfying $2^{p-1} \equiv 1 \pmod{p^2}$ are $1093, 3511$; the primes satisfying $3^{p-1} \equiv 1 \pmod{p^2}$ are $11, 1006003$. Their intersection is empty. Theorem 4 now applies, with no bound on h other than the good-prime restriction. □

4 General negative integer centres

Proposition 6. *Fix $m \geq 2$, even $E \geq 6$, even $h \geq 2$, and $p > m$. If for $j = 0, \ldots, m-1$ one has $-(n_j + m)! S_{n_j}(-m) \equiv 0 \pmod{p^k}$, then for each $1 \leq a \leq m$,*

$$p^k \;\Big|\; \prod_{\substack{1 \leq b \leq m \\ b \neq a}} (a^h - b^h).$$

Consequently all possible prime divisors for a fixed pair (m, h) divide an explicit integer independent of E, and the p-adic depth obeys

$$k \leq \min_{1 \leq a \leq m} \sum_{\substack{1 \leq b \leq m \\ b \neq a}} v_p(a^h - b^h).$$

Proof. Define $\lambda_a = a^h$, $U_a = (-1)^{m-a}\binom{m}{a} a^{E+m-1}$, and $Z_j = \sum_a U_a \lambda_a^j$ using (1). For fixed a, write $P_a(T) = \prod_{b \neq a}(T - \lambda_b) = \sum_{j=0}^{m-1} c_j T^j \in \mathbb{Z}[T]$. Then

$$\sum_{j=0}^{m-1} c_j Z_j = \sum_{b=1}^{m} U_b P_a(\lambda_b) = U_a \prod_{b \neq a} (\lambda_a - \lambda_b).$$

The left side is divisible by p^k, whereas U_a is a p-adic unit because $p > m$. Divide by U_a. □

5 The first shifted Hensel jet at -3

This section concerns a common root $X \equiv -3 \pmod{p}$, which is allowed to shift at higher precision. It does *not* identify or restrict common roots in other residue disks. Let

$$F_j(X) = -(n_j + 3)! S_{n_j}(X), \qquad P_j = F_j'(-3), \quad j = 0, 1, 2.$$

At a good prime $p > 2n_2$ these have p-integral polynomial coefficients and $F_j(-3) = 3C_j$. In particular their Taylor expansions at -3 make sense modulo p^2.

Proposition 7 (Finite-prime obstruction for shifted triples)**.** *Suppose E, h are even, $E \geq 6, h \geq 2$, and $p > 2n_2$ is prime. If there is an $X \equiv -3 \pmod{p}$ such that $q_{n_j}(X) \equiv 0 \pmod{p}$ for all $j = 0, 1, 2$, then*

$$p \mid \gcd(2^h - 1, 3^h - 1).$$

Consequently for each fixed gap h there are only finitely many eligible primes and triples (p, E, h) with such a shifted residue disk, regardless of the precision to which it lifts.

Proof. Reduction $X \equiv -3 \pmod{p}$ and the good-prime normalizer give $C_0 \equiv C_1 \equiv C_2 \equiv 0 \pmod{p}$. Theorem 2 with $k = 1$ gives $2^h \equiv 3^h \equiv 1 \pmod{p}$. For fixed h, the gcd is a fixed nonzero integer. Each eligible p limits E by $2(E + 2h - 1) < p$. □

Theorem 8 (First-order shift-curvature obstruction)**.** *Assume $C_0 \equiv C_1 \equiv C_2 \equiv 0 \pmod{p}$, where $p > 2n_2$. If an integer u satisfies*

$$q_{n_j}(-3 + pu) \equiv 0 \pmod{p^2} \qquad (j = 0, 1, 2),$$

then

$$\boxed{u(P_2 - 2P_1 + P_0) \equiv 0 \pmod{p}.} \tag{5}$$

In particular, if $P_2 - 2P_1 + P_0$ is a unit modulo p, every such common lift must have $u \equiv 0 \pmod{p}$.

Proof. Theorem 2 with $k = 1$ shows that $x = 2^h - 1$ and $y = 3^h - 1$ are both divisible by p. Thus the exact second difference at the fixed centre is

$$F_2(-3) - 2F_1(-3) + F_0(-3) = 3(My^2 - Lx^2) \equiv 0 \pmod{p^2}.$$

Taylor expansion modulo p^2 gives $F_j(-3 + pu) \equiv F_j(-3) + puP_j \pmod{p^2}$. Subtract twice the middle congruence from the first and third. The three shifted values vanish modulo p^2, so $pu(P_2 - 2P_1 + P_0) \equiv 0 \pmod{p^2}$. Cancel p to obtain (5). □

Theorem 9 (Exact first-digit compatibility for shifted triples)**.** *Retain the hypotheses of Theorem 8. Define residues in $\mathbb{F}_p$ by*

$$A = F_0(-3)/p, \qquad B = (F_1(-3) - F_0(-3))/p, \qquad d_j = F_j'(-3) \quad (j = 0, 1, 2).$$

These quotients are integral because all three fixed-centre values vanish modulo p. Put $e = d_1 - d_0$ and $D = d_2 - 2d_1 + d_0$. There exists a common lifted root $X = -3 + pu$ modulo p^2 if and only if the following finite-field conditions hold:

1. *If $D \neq 0$, then $A = B = 0$. In this case $u = 0$ is the only possible first digit.*
2. *If $D = 0$ but $(d_0, e) \neq (0, 0)$, then $Ae - Bd_0 = 0$. In this case a unique u satisfies $A + ud_0 = B + ue = 0$.*
3. *If $D = 0$ and $(d_0, e) = (0, 0)$, then $A = B = 0$. In this case every $u \in \mathbb{F}_p$ is a common lift at this precision.*

The conditions concern only the specified disk $X \equiv -3 \pmod{p}$.

Proof. The fixed-centre triple modulo p implies $2^h \equiv 3^h \equiv 1 \pmod{p}$, so the exact difference identity gives

$$F_2(-3) - 2F_1(-3) + F_0(-3) \equiv 0 \pmod{p^2}.$$

Therefore $F_j(-3)/p = A + jB$ in $\mathbb{F}_p$ for $j = 0, 1, 2$. Taylor expansion around -3 gives

$$F_j(-3 + pu)/p \equiv A + jB + ud_j \pmod{p}.$$

The second difference of the three right sides is uD. When $D \neq 0$ this forces $u = 0$, leaving $A = B = 0$. When $D = 0$ the derivative vector is affine, $d_j = d_0 + je$, and the three equations reduce to $A + ud_0 = 0$ and $B + ue = 0$. An elementary two-dimensional linear-dependence test gives the remaining two alternatives, including the degenerate zero vector. □

Remark 10. *The exact first-digit test combines a CCR-specific affine fixed-centre identity with elementary linear Hensel algebra. It is not a depth-four existence theorem, and we do not assert that the underlying linear algebra is a new technique.*

Proposition 11 (The $p = 431$ triple is obstructed at depth two)**.** *For* $(p, E, h) = (431, 42, 86)$ *and* $(n_0, n_1, n_2) = (41, 127, 213)$*, there is no common root* $X \equiv -3 \pmod{431}$ *of the three quotient polynomials modulo* 431^2*.*

Proof. Exact finite-field coefficient arithmetic gives

$$(P_0, P_1, P_2) \equiv (62, 191, 86) \pmod{431}, \qquad P_2 - 2P_1 + P_0 \equiv 197 \not\equiv 0 \pmod{431}.$$

In the notation of Theorem 9, the fixed-centre first quotients are $(A, B) = (274, 117)$ modulo 431. The curvature $D = 197$ is a unit, but $A \neq 0$; hence the exact first-digit compatibility criterion excludes a common lift. In fact the individual required digits are 79, 359, and 385. These residues can be obtained by exact coefficient extraction from $H(t)^3$ and $-H(t)^3 \log H(t)$ with $H(t) = (1 - e^{-t})/t$, using the reciprocal-series identity and the coefficient recurrence derived from $A(t)f'(t) = XA'(t)f(t)$ for $A(t) = t/(1 - e^{-t})$. The values displayed above give the finite-field certificate needed for the stated first-digit obstruction. □

Remark 12 (Scope and originality)**.** *The proof of Theorem 8 uses elementary Taylor expansion and a special fixed-centre second-difference identity. It is a useful* specialized consequence*, not a claim that Hensel's lemma or the jet method is new. It does not constrain an arbitrary pair* (q_r, q_s) *in a residue disk other than* -3*.*

6 All-prime finite-gap exclusion of shifted triples

The fixed-centre classification and the first-digit shift criterion combine to exclude *shifted* triples uniformly over all primes within a nontrivial range of index gaps. This is an explicitly certified finite-gap theorem; it is not a universal statement over unbounded h.

Theorem 13 (Complete $h \leq 120$ shifted-triple exclusion)**.** *Let* $E \geq 6$ *and* $2 \leq h \leq 120$ *be even, put* $n_j = E + jh - 1$ $(j = 0, 1, 2)$*, and let* $p > 2n_2$ *be a prime. There is* no *integer* $X \equiv -3 \pmod{p}$ *for which*

$$q_{n_0}(X) \equiv q_{n_1}(X) \equiv q_{n_2}(X) \equiv 0 \pmod{p^2}.$$

The conclusion has no upper bound on p*. In particular no such disk has simultaneous depth* $k \geq 2$ *for three equally spaced orders in this gap range.*

Proof. If the displayed congruences held, the three values at -3 would vanish modulo p. By Theorem 2,

$$p \mid \gcd(2^h - 1, 3^h - 1), \qquad p > 4h + 10.$$

A self-contained deterministic certificate for the 60 even gaps uses a shorter complete test than factoring all 60 gcd values: for each gap h, set $g_h = \gcd(2^h - 1, 3^h - 1)$ and divide out *every prime* $\ell \leq 4h + 10$ to its full multiplicity. If the remainder is 1, no prime factor of g_h exceeds the necessary goodness threshold; if the remainder is larger, all its prime factors exceed the threshold. Eratosthenes sieving up to 490 and exact integer division give remainder 1 for 59 gaps and remainder 431 for $h = 86$. Trial division through $\lfloor\sqrt{431}\rfloor$ confirms that 431 is prime. Thus the *only* pair surviving $p > 4h + 10$ is

$$(h, p) = (86, 431).$$

The goodness inequality then leaves finitely many even E; checking $3 \cdot 3^E - 4 \cdot 2^E + 1 \equiv 0 \pmod{431}$ gives only $E = 42$. Consequently the *only* first-level triple in this entire range is $(p; n_0, n_1, n_2) = (431; 41, 127, 213)$.

At this triple, all three roots are simple modulo 431 in the -3 disk. Independent modular series coefficient arithmetic gives the unique first Hensel digits of the three polynomial roots:

$$u_{41} = 79, \qquad u_{127} = 359, \qquad u_{213} = 385.$$

Because these residues are pairwise distinct, no shared u exists with $X = -3 + 431u \pmod{431^2}$. Thus the remaining case is also excluded. This finishes the finite certificate: for each of the 60 even gaps, form the exact integer gcd and remove all prime powers at or below $4h + 10$; the sole nonunit remainder is 431 at $h = 86$. The good-prime inequality then reduces the surviving case to $E = 42$, and the displayed distinct first Hensel digits exclude its only possible common lift. The certificate requires exact integer and finite-field computations, but no further assumption about an untested prime range. □

Remark 14 (Computational scope and logical strength)**.** *The certificate is finite* in the gap*, not in the prime: a fixed nonzero integer* $\gcd(2^h - 1, 3^h - 1)$ *forces all possible primes into a finite list for each* h*. Certifying every gap up to* 120 *therefore yields a genuine all-prime result in that range, unlike an exhaustive search over primes with a fixed numeric bound. No assertion is made for a different residue disk or for a pair of orders alone. This result combines earlier exact project identities and computation; the calculation itself is part of the supporting finite certificate.*

7 Finite verification, certification and relation to earlier results

The exact computations play two different roles in the results above. For Corollary 5, a complete sieve of the 664,577 primes $5 \le p \le 10^7$ finds only $1093, 3511$ for the base-2 Wieferich condition and only $11, 1006003$ for the base-3 condition, with empty intersection. This is a *prime-bounded* exclusion and does not assert that no larger simultaneous Wieferich prime exists.

Theorem 13 instead has a *finite-gap, all-prime* computer-assisted component: for each of the 60 even gaps $2, 4, \ldots, 120$, exact gcd and prime-stripping calculations give remainder 1 in 59 cases and 431 only for $h = 86$. At that gap, the good-prime bound and the exact fixed-centre congruence leave only $E = 42$, with indices $(41, 127, 213)$. The three required first Hensel digits are $(79, 359, 385)$ modulo 431, hence no common lifted root exists. This is a finite and fully specified arithmetic certificate for the theorem's unlimited-prime conclusion *within that gap range*; the theorem is not being presented as a computation-free symbolic proof.

Additional bounded consistency checks include 30 low-degree exact-rational instances of (1), 180 interpolation identities, an exhaustive $5^5 = 3125$ five-field-instance check of the affine-jet criterion, and an independent exact modular-coefficient computation of the $(p, E, h) = (431, 42, 86)$ residues. These finite checks corroborate the stated identities and the exceptional certificate; they do not replace the symbolic arguments in the general results or justify extending any theorem beyond its hypotheses. No verification program, data archive, or supplementary code is claimed to be distributed with this article.

Classical finite-difference and Nörlund methods appear in Adelberg [7, 8, 9], and general Bernoulli–Stirling congruences in Young [10]. Results on prime divisors of resultants, relevant to broader collision searches, were proved by Gómez-Pérez et al. [11]. The algebraic principles used here are elementary and established; our conclusions are the stated fixed-centre and finite-gap results, not new general Hensel or finite-difference methods.

The previously prepared CCR works [1, 2, 3, 4] provide the chronological-carry and Stirling-polynomial provenance. A separate existing article by the same author [6] studies single-site-to-prefix rigidity for two *prime-indexed* quotient polynomials, whereas this article treats three equal-gap indices, simultaneous fixed-centre depth and the first shifted lift. Both use classical Stirling identities and the same good-prime normalization; neither article treats its shared preliminary lemmas as an independent new contribution.

The exact first-digit criterion and the finite-gap exclusion here cover three equal-gap orders only in the residue disk $X \equiv -3 \pmod{p}$; it does not generalize the Wieferich conclusion to shifted pairs. The known depth-three example of two distinct orders at a shifted half-integer centre lies outside these hypotheses. The fixed-centre and triple-shifted statements hold precisely in the residue disks and order configurations specified in their hypotheses.

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